HDOC DAY-9 REPORT

Menu-Driven C Programming Activities

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Batch	15
Activity	HDOC Day-9

Prepared for academic submission

Question 1: Swap Two Numbers

Problem Statement

Write a menu-driven C program to input two numbers and swap them using a third variable or without using a third variable. Display the values before and after swapping.

Algorithm

1. Start
2. Input two numbers a and b
3. Display menu: 1) With third variable 2) Without third variable
4. Read choice
5. If choice = 1: temp = a, a = b, b = temp
6. If choice = 2: a = a + b, b = a - b, a = a - b
7. Display swapped values
8. Stop

Pseudocode

BEGIN
INPUT a, b
DISPLAY menu options
INPUT choice
IF choice = 1 THEN use temp to swap
ELSE IF choice = 2 THEN swap using arithmetic
ELSE DISPLAY invalid choice
DISPLAY a, b
END

Flowchart

Start

↓

Input two numbers a and b

↓

Display menu: 1) With third variable 2) Without third variable

↓

Read choice

↓

If choice = 1: temp = a, a = b, b = temp

↓

If choice = 2: a = a + b, b = a - b, a = a - b

↓

Display swapped values

↓

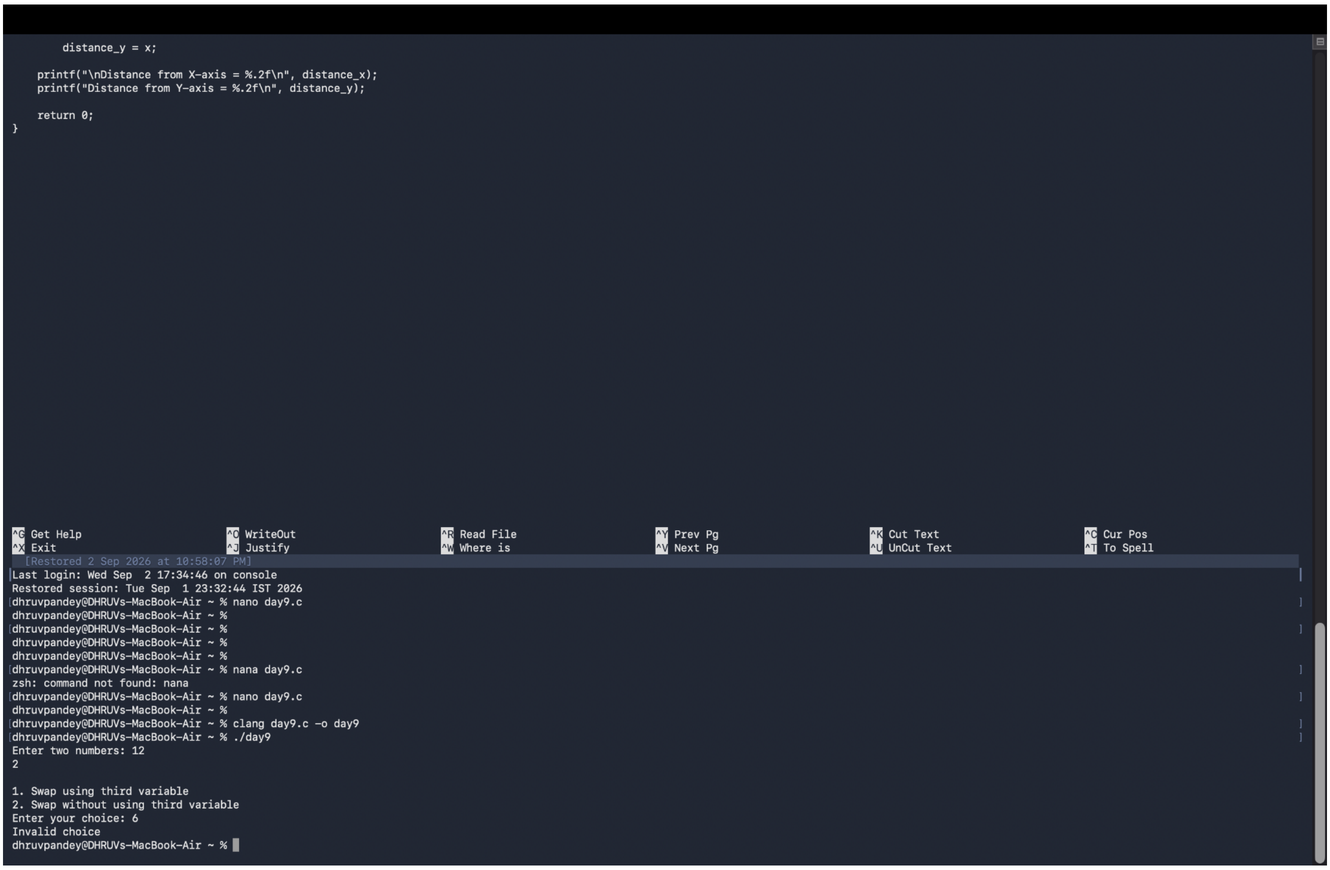
Stop

Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	a=12, b=2, choice=1	a=2, b=12	a=2, b=12	Pass
2	a=12, b=2, choice=2	a=2, b=12	a=2, b=12	Pass
3	a=12, b=2, choice=6	Invalid choice	Invalid choice	Pass

Question 1: Swap Two Numbers - Code / Execution Screenshots

Execution screenshot provided:



Question 2: Sum of Natural Numbers / Sum of Squares

Problem Statement

Write a menu-driven C program to find either the sum of the first n natural numbers or the sum of squares of the first n natural numbers.

Algorithm

1. Start
2. Input n
3. Display menu: 1) Sum of first n natural numbers 2) Sum of squares
4. Read choice
5. If choice = 1: add 1 through n
6. If choice = 2: add 1^2 through n^2
7. Display result
8. Stop

Pseudocode

BEGIN
INPUT n
DISPLAY menu
INPUT choice
SET sum = 0
IF choice = 1 THEN repeat i=1 to n: sum = sum + i
ELSE IF choice = 2 THEN repeat i=1 to n: sum = sum + i*i
ELSE DISPLAY invalid choice
DISPLAY sum
END

Flowchart

Start

↓

Input n

↓

Display menu: 1) Sum of first n natural numbers 2) Sum of squares

↓

Read choice

↓

If choice = 1: add 1 through n

↓

If choice = 2: add 1^2 through n^2

↓

Display result

↓

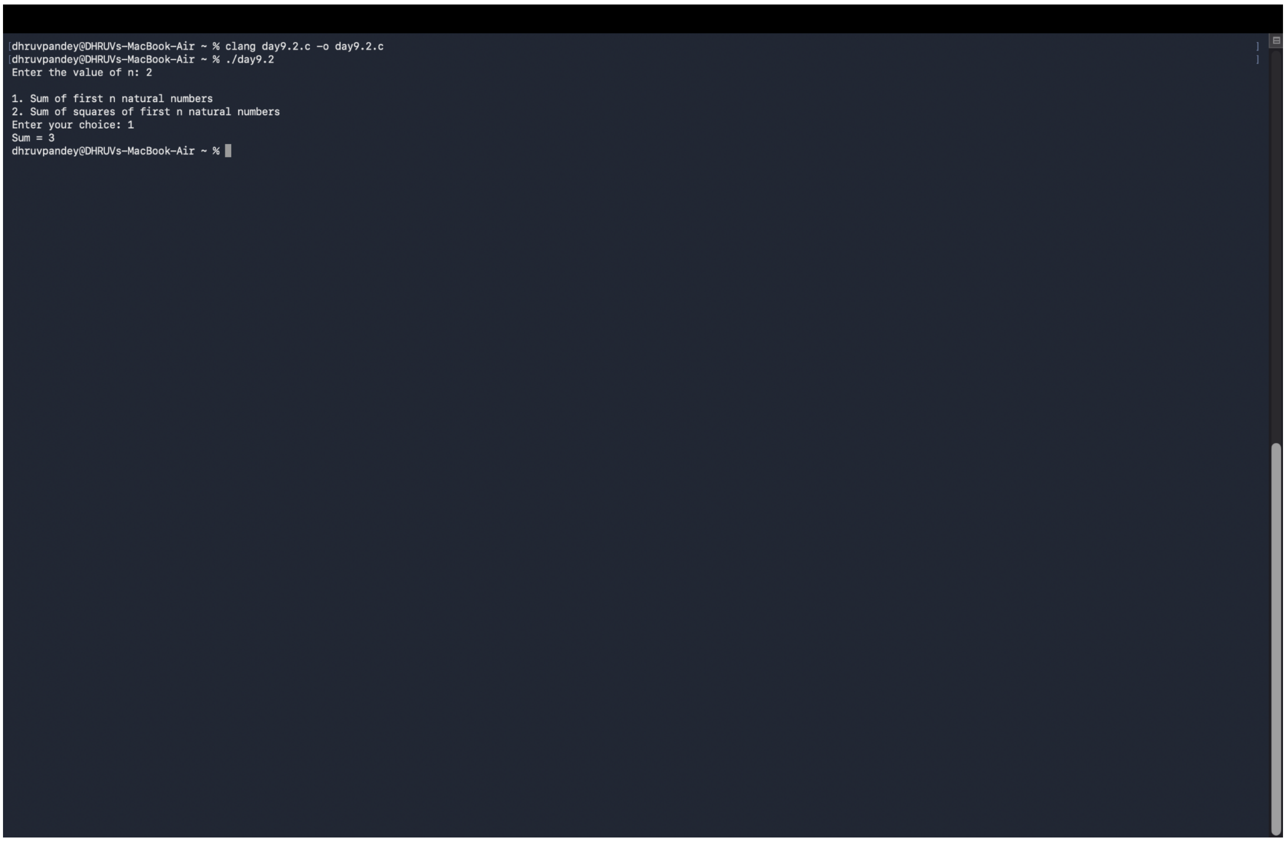
Stop

Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n=2, choice=1	Sum = 3	Sum = 3	Pass
2	n=3, choice=2	Sum of squares = 14	Sum of squares = 14	Pass
3	n=12, choice=12	Invalid choice	Invalid choice	Pass

Question 2: Sum of Natural Numbers / Sum of Squares - Code / Execution Screenshots

Execution screenshots provided:



Question 3: Interest Calculations

Problem Statement

Write a menu-driven C program to calculate simple interest, compound amount, compound interest, or total payable amount for a given principal, rate, and time.

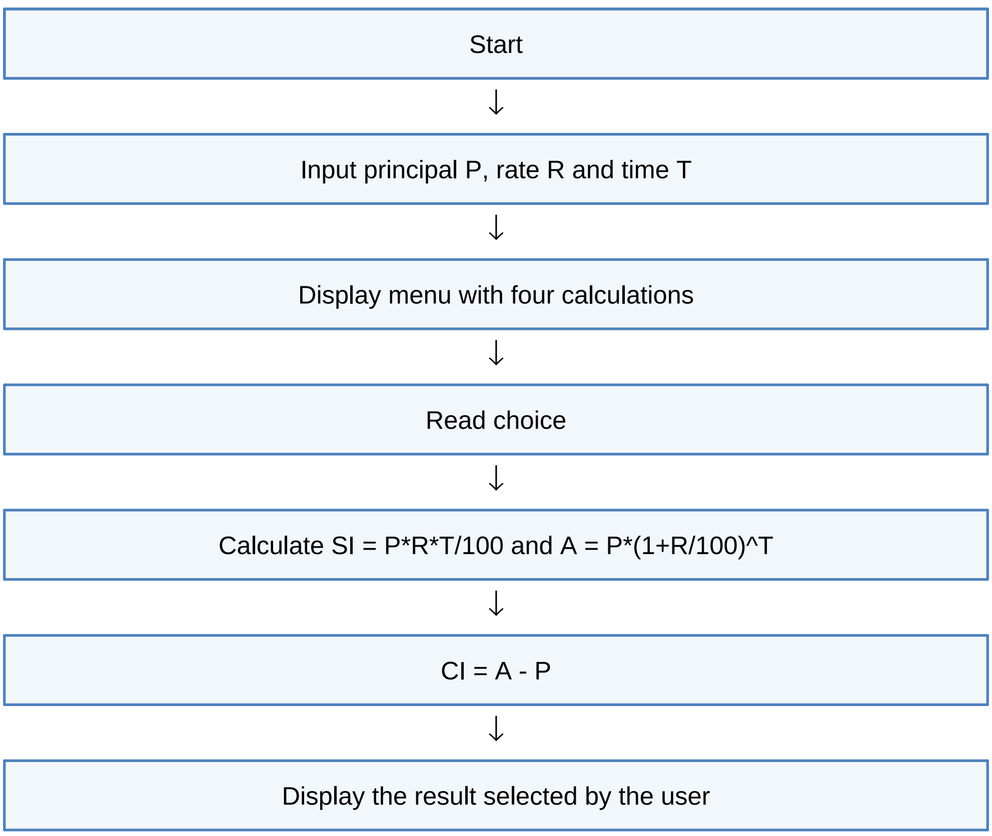
Algorithm

1. Start
2. Input principal P, rate R and time T
3. Display menu with four calculations
4. Read choice
5. Calculate SI = P*R*T/100 and A = P*(1+R/100)^T
6. CI = A - P
7. Display the result selected by the user
8. Stop

Pseudocode

BEGIN
INPUT P, R, T
DISPLAY 1. Simple Interest, 2. Compound Amount, 3. Compound Interest, 4. Total Payable
INPUT choice
SI = (P*R*T)/100
A = P*(1+R/100)^T
CI = A-P
IF choice=1 DISPLAY SI; choice=2 DISPLAY A; choice=3 DISPLAY CI; choice=4 DISPLAY P+SI
END

Flowchart



↓

Stop

Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	P=1000, R=10, T=2, choice=1	Simple Interest = 200.00	200.00	Pass
2	P=1000, R=10, T=2, choice=2	Compound Amount = 1210.00	1210.00	Pass
3	P=1000, R=10, T=2, choice=3	Compound Interest = 210.00	210.00	Pass
4	P=1000, R=10, T=2, choice=4	Total Payable Amount = 1200.00	1200.00	Pass

Question 3: Interest Calculations - Code / Execution Screenshots

Code screenshots provided (no typed code included):

```
  UW PICO 5.09                                  File: day9.4.c                                  Modified

#include <stdio.h>
#include <math.h>

int main()
{
    double principal, rate, time;
    double simpleInterest, compoundAmount;
    double compoundInterest, totalPayable;
    int choice;

    printf("Enter principal amount: ");
    scanf("%lf", &principal);

    printf("Enter rate of interest: ");
    scanf("%lf", &rate);

    printf("Enter time in years: ");
    scanf("%lf", &time);

    printf("\n1. Simple Interest");
    printf("\n2. Compound Amount");
    printf("\n3. Compound Interest");
    printf("\n4. Total Payable Amount");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    simpleInterest = (principal * rate * time) / 100;

    compoundAmount =
        principal * pow((1 + rate / 100), time);

    compoundInterest = compoundAmount - principal;

    switch(choice)
    {
        case 1:
            printf("Simple Interest = %.2lf\n", simpleInterest);
            break;

        case 2:
            printf("Compound Amount = %.2lf\n", compoundAmount);
            break;

        case 3:
            printf("Compound Interest = %.2lf\n", compoundInterest);
            break;

        case 4:
            totalPayable = principal + simpleInterest;
            printf("Total Payable Amount = %.2lf\n", totalPayable);
            break;

        default:
            printf("Invalid choice\n");
    }
File Name to write : day9.4.c
^G Get Help                        ^T  To Files
^C Cancel                          TAB Complete
```

```
  UW PICO 5.09                                  File: day9.4.c                                  Modified

#include <stdio.h>
#include <math.h>

int main()
{
    double principal, rate, time;
    double simpleInterest, compoundAmount;
    double compoundInterest, totalPayable;
    int choice;

    printf("Enter principal amount: ");
    scanf("%lf", &principal);

    printf("Enter rate of interest: ");
    scanf("%lf", &rate);

    printf("Enter time in years: ");
    scanf("%lf", &time);

    printf("\n1. Simple Interest");
    printf("\n2. Compound Amount");
    printf("\n3. Compound Interest");

  printf("\n4. Total Payable Amount");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    simpleInterest = (principal * rate * time) / 100;

    compoundAmount =
        principal * pow((1 + rate / 100), time);

    compoundInterest = compoundAmount - principal;

    switch(choice)
    {
        case 1:
            printf("Simple Interest = %.2lf\n", simpleInterest);
            break;

        case 2:
            printf("Compound Amount = %.2lf\n", compoundAmount);
            break;

        case 3:
            printf("Compound Interest = %.2lf\n", compoundInterest);
            break;

        case 4:
            totalPayable = principal + simpleInterest;
            printf("Total Payable Amount = %.2lf\n", totalPayable);
            break;

        default:
            printf("Invalid choice\n");
                                          [ Unknown Command:  ]
^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Pg       ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify       ^W Where is      ^V Next Pg       ^U UnCut Text    ^T To Spell
```

Question 3: Interest Calculations - Code Screenshot Continued

```
  UW PICO 5.09                                                                          File: day9.4.c

    compoundAmount =
        principal * pow((1 + rate / 100), time);

    compoundInterest = compoundAmount - principal;

    switch(choice)
    {
        case 1:
            printf("Simple Interest = %.2lf\n", simpleInterest);
            break;

        case 2:
            printf("Compound Amount = %.2lf\n", compoundAmount);
            break;

        case 3:
            printf("Compound Interest = %.2lf\n", compoundInterest);
            break;

        case 4:
            totalPayable = principal + simpleInterest;
            printf("Total Payable Amount = %.2lf\n", totalPayable);
            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}

^G Get Help     ^O WriteOut     ^R Read File     ^Y Prev Pg     ^K Cut Text       ^C Cur Pos
^X Exit         ^J Justify      ^W Where is      ^V Next Pg     ^U UnCut Text     ^T To Spell
```

Conclusion

The Day-9 activities demonstrate the use of menu-driven programming, switch-case selection, arithmetic operations, loops, and mathematical formulas in C. The programs were prepared and tested using representative test cases.